CLEANING, THERMAL DISINFECTION, AND STERILIZATION

These cleaning, disinfection, and sterilization instructions have been validated for preparing reusable DePuy Synthes instruments for reuse. It is the responsibility of the end user to ensure that the cleaning, disinfection, and sterilization is performed using appropriate equipment, materials, and personnel to achieve the desired result. Any deviation from these instructions should be evaluated for effectiveness and potential adverse consequences.

General considerations for processing

- Observe point of use and transport procedures as described below.
- Highly alkaline conditions (pH>10) can damage components or device.
- Complete each procedure: preparation for cleaning, manual cleaning or automated cleaning, disinfection, and sterilization when reprocessing the instruments.
- Personnel trained in the appropriate reprocessing techniques and safety should complete all reprocessing. Wear appropriate protective equipment (gloves, eye protection, etc.) when reprocessing any medical device.

Point of use care

- The drying of gross soil (blood, tissue and/or debris) on devices following surgical use should be avoided. It is preferred that gross soil is removed from devices following use and in preparation for transportation to a processing area. Gross soil can be removed using sponges, cloths, or soft brushes. Water and/or cleaning detergents (labelled for use on medical devices) may be used.
- Do not use saline, environmental disinfection (including chlorine solutions) or surgical antiseptics (such as iodine- or chlorhexidine-containing products). Flush all lumens, blind holes, small clearances, and moving and intricate parts with water (or detergent solution) to prevent the drying of soil and/or debris.
- If gross soil cannot be removed at the point of use, the devices should be transported to prevent drying (e.g., covered with a towel dampened with purified water) and cleaned as soon as possible at a designated processing area.

Transport to processing area

Surgically used devices may be considered bio-hazardous and should be safely transported to a designated processing area in accordance with local policies. Complete the cleaning procedures for the instruments within 30 minutes following their use.

Preparation for cleaning

- It is recommended that devices should be reprocessed as soon as is reasonably practical following surgical use.
- Instruments must be cleaned separately from instrument trays and cases.
- Care should be taken in the handling and cleaning of sharp devices. These are recommended to be cleaned separately to reduce risks of injury.
- Disassemble the instruments according to the following instructions:

1. Disconnect the light cable from the lightguide post.
2. Disconnect the camera-coupling device from the eyepiece.
3. Remove any light cable adaptors from the endoscope.
4. Disassemble the endoscope from any devices or compatible devices used during the procedure.
5. Disassemble the sheath including the stopcocks according to Figure 2.
6. Perform either the manual cleaning procedure or the automated cleaning procedure described below.

MANUAL CLEANING

Perform the following steps at the point of central reprocessing:

1. Rinse the instruments under cold running tap water for a minimum of three (3) minutes. Brush all surfaces and actuate moving parts during the rinse.

2. Prepare a neutral (pH 7-9) detergent (cleaning) solution (which may or may not include enzymes) in accordance to the detergent manufacturer's instructions. Use a fresh mixture of cleaning solution for each set of instruments when cleaning.

   NOTE: The cleaning solution may contain enzymes. Aluminum-safe alkaline cleaners can be used but can vary in material compatibility overtime based on their formulation. Material compatibility should be confirmed with the detergent manufacturer.
3. Immerse the instruments completely in the detergent solution for a minimum of 15 minutes.
4. Soak for five (5) minutes and then use soft bristle brushes to remove any debris or soil from the instruments while they are submerged. Brush for a minimum of 3 minutes. Ensure that the brushes can access the hard-to-reach areas such as cannulations, cracks, crevices, and threads. Remove all visible debris.
5. Fill a syringe with the detergent and use to flush cannulations, crevices, and threads while submerged.
6. Repeat the brushing step (step 3) and the flushing step (step 4) an additional 2 times.
7. Check for remaining debris and remove as needed.
8. For the sheath only, completely submerge the devices in an ultrasonic bath. Carry out an ultrasonic second soaking bath in a neutral pH detergent solution for 10 minutes using an ultrasonic bath (refer to cleaning agent solution manufacturer's instructions for immersion time and temperature).

   NOTE: Ultrasonic cleaning is only effective if the surface to be cleaned is immersed in the cleaning solution. Air pockets will decrease the efficacy of ultrasonic cleaning.
9. Rinse the instruments under cold running tap water for at least three (3) minutes while brushing with a soft bristle brush during the rinse.
10. For the final rinsing processes, soak the devices for a minimum of three (3) minutes in purified (e.g., deionized, reverse osmosis, distilled, or critical water), per AAMI TIR 34. Dry with a lint-free soft cloth.

AUTOMATED CLEANING

Equipment required: Automated washing shall be conducted in a validated washer-disinfector in compliance to ISO 15883-1 and-2, or to an equivalent standard. Automated washing can be included as part of a validated washing, disinfection, and/or drying cycle in accordance with manufacturer's instructions.

1. Manually clean the instruments by immersion prior to automated cleaning (reference manual cleaning section).
2. Place the endoscope and its light cable adaptors into a separate mesh basket. Ensure that the endoscope and light cable adaptors do not touch each other when placed into the basket. Place a mesh screen over the basket to contain the endoscope and light cable adaptors in the basket during processing.
3. Clean in a validated washer disinfector using the "INSTRUMENTS" cycle and a pH neutral cleaning agent intended for use in automated cleaning as shown in Table 2. **NOTE:** Enzol® (1oz/gal) and Prolystica 2X Neutral® (1/8 oz/gal) were used for the validation.

Table 2: Automatic cleaning parameters.

Phase	Recirculation time set points	Water temperature	Detergent type
Pre-wash	2 minutes	Cold tap water	Not applicable
Enzyme wash	1 minute	Hot tap water	Enzymatic detergent
Wash 1	2 minutes	65°C (set point)	Neutral pH instrument cleaner
Rinse 1	15 seconds	Hot tap water	Not applicable
Dry phase	15 minutes	66°C (set point)	Not applicable

INSPECTION AFTER CLEANING

After either manual cleaning or automated cleaning, visually inspect the instruments to verify the absence of visible soil, stains, and debris. If soil is present after cleaning, repeat cleaning procedure.

Thermal Disinfection

Thermal disinfection is recommended to render devices safe for handling prior to steam sterilization. Thermal disinfection should be conducted in a washer-disinfector compliant to ISO 15883-1 and-2, or to an equivalent standard. Thermal disinfection in the washer-disinfector shall be validated to provide an A_0 of at least 600 (e.g., 90°C (194°F) for 1 min). Higher levels of A_0 can be achieved by increasing the exposure time and temperature (e.g., A_0 of 3000 at >90°C (194°F) for 5 min, in accordance with local requirements). Load the device components in the washer-disinfector in accordance with manufacturer's instructions, ensuring that the devices and lumens can drain freely. Lumened devices should be placed in a vertical position. If this is not possible due to space limitations within the washer-disinfector, use an irrigating rack or load carrier with connections designed to ensure an adequate flow of process fluids to the lumen or cannulation of the device if provided.

The following automated cycles are examples of validated cycles (Table 3):

Table 3: Examples of automated validated cycles.

Phase	Recirculation time (mins)	Water temp	Water type
Thermal disinfection	1	> 90°C (194°F)	Critical water (RO, deionized or distilled water)
Thermal disinfection	5	> 90°C (194°F)	Critical water (RO, deionized or distilled water)

STERRAD™ sterilization: endoscope and light adaptors only

CAUTION: The sheaths are not validated for sterilization using STERRAD.

1. Ensure the endoscope and light cable adaptors are disassembled before sterilization.
2. The endoscope and light cable adaptors can be sterilized in the STERRAD™ 100S system, and in the STERRAD™ 100NX, and STERRAD™ NX sterilization systems using the STANDARD cycle.
3. Clean and thoroughly dry the endoscope and light cable adaptors per the section titled "CLEANING, THERMAL DISINFECTION, AND STERILIZATION."
4. Place the endoscope and light cable adaptors into the appropriate location within the Mitek Sport Medicine sterilization tray that is compatible with the STERRAD 100S, NX, and 100NX sterilization systems. Package trays or instruments with a barrier wrap material in accordance with local procedures, using standardized wrapping techniques such as those described in ANSI/AAMI 79. In the United States, use an FDA-cleared sterilization wrap. Include a STERRAD indicator strip in the wrapped package. Run the STERRAD machine per the manufacturer's instructions.

Autoclave (steam) sterilization

CAUTION: Do not use an "immediate use or flash" cycle to sterilize the instruments.

1. Before assembly of the sheath for sterilization, make sure that all O-rings are in place and are not damaged. Remove and replace any O-rings that show any sign of damage or wear. Use a water-soluble lubricant in accordance with the lubricant manufacturer's instructions to lubricate parts as necessary. Remove any excess lubricant. Sterilize the sheath with the stopcocks in the open position.
2. Sterilize the endoscope and light cable adaptors disassembled.
3. If the instruments come with a protective transport sleeve, do not sterilize the endoscopic compatible devices with the protective transport sleeve.